ตัวอย่าง · ตัดข้อมูลลูกค้าและราคาออกแล้ว · Sample: customer details and prices removed

Solar ESS Co. Ltd.
53 Moo 2, Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel. 063 151 6694, 089 632 5829 · www.solaress.org

Hybrid Solar + Battery Package

18.9 kWp solar · 20 kW 3-phase hybrid inverter · 48 kWh battery · generator-ready

18.9 kWp	20 kW	48 kWh	~70 kWh
30 × 630 W panels	3-phase hybrid inverter	3 × 16 kWh LFP battery	solar per sunny day

Who it is for. Homes, resorts, shops and small businesses on a 3-phase supply with a large evening load or frequent power cuts. The panels run the daytime load and charge the battery; the battery carries the evening and night; the grid or a generator covers the rest. If PEA power is cut, the hybrid inverter keeps the backed-up circuits running from the battery and solar. An on-grid-only system cannot do this: it switches off whenever PEA power is off.

What is included

Item	Specification	Qty
Solar panels	JA Solar JAM66D45-630/LB, 630 W n-type bifacial, 3 strings of 10	30 pcs
Hybrid inverter	Deye SUN-20K-SG05LP3-EU-SM2, 20 kW 3-phase, 48 V battery, 2 MPPT, generator port	1 set
Battery	16 kWh LFP, 51.2 V 314–320 Ah, 6,000 cycles, CAN link to the inverter	3 units
Smart meter	3-phase meter with CTs for zero export and load monitoring	1 set
Mounting	Aluminium rails, L-feet or hooks to suit the roof, panel clamps, grounding clips, waterproofing	1 lot
DC combiner box	KJL steel box; per string: Suntree DC breaker 2P 800 V 32 A and DC SPD 2P 800 V (Type II)	3 strings
AC distribution box	CHINT NXM-63S 4P 40 A MCCB for inverter output and for the generator input; Suntree SUP1H-40 4P AC SPD; earth bar	1 set
Battery DC box	Copper busbars on insulators, clear protective cover, main battery DC breaker 2P 400 A class	1 set
Battery cables	50 mm² per battery (equal lengths); 2 × 70 mm² per pole from busbar to inverter	1 lot
Cabling and conduit	LINK PV cable, Yazaki copper AC cable, IMC/EMT conduit or cable tray, MC4 connectors	1 lot
Earthing	Copper earth rods, main earth conductor, equipotential bonding of frames, rails and boxes	1 lot
Services	Installation by certified electricians, testing to IEC 62446, PEA grid-connection documents with an engineer-certified SLD, commissioning and app setup	1 lot

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Package price

Item	Description	Unit	Qty	Unit price	Amount (THB)
PV system	18.9 kWp solar with 20 kW hybrid inverter, smart meter, mounting, DC/AC boxes, cabling, earthing, installation, testing and PEA documents	kWp	18.9	–	–
Battery	16 kWh LFP battery, 51.2 V	unit	3	–	–
Battery cable 50 mm²	Battery to busbar, equal lengths	m	24	–	–
Battery cable 70 mm²	Busbar to inverter, 2 per pole	m	12	–	–
Battery DC box	Copper busbars, insulators, cover and main 2P 400 A-class DC breaker	set	1	–	–
Battery installation	Placement on plinth, cabling, BMS communication set-up and testing	lot	1	–	–
Total before VAT					–
VAT 7%					–
Package total					–

Labour within the price is subject to 3% withholding tax, deducted by the customer at payment with a withholding-tax certificate issued to the company.

Options

Option	What it adds	Price (THB, excl. VAT)
Deye SE-F16 batteries	Same-brand battery as the inverter (16 kWh, 314 Ah) for one-stop warranty support; replaces the standard battery	–
4th string, +6.3 kWp	10 more panels on the spare MPPT capacity: 25.2 kWp in total, about +25 kWh per sunny day, with its own DC breaker and SPD	On request
Generator connection	Connection point and ATS to the inverter generator port. The generator itself is supplied, installed and warranted by the generator company	On request

Expected performance

Annual solar generation	about 26,460 kWh
Used on site (with battery, about 90%)	about 23,814 kWh
Bill savings at 4.5 THB/kWh	about 107,163 THB per year
Simple payback	about 6.5 years
Battery runtime, 3 kW night load / 5 kW night load	about 14 h / 8–9 h (43 kWh usable)

Estimates for Chiang Rai (about 1,400 kWh per kWp per year, panels facing south at a good tilt). Actual results depend on roof direction, shading, weather and how much power is used in the daytime and evening. Selling surplus power back to PEA is not available; the system runs with zero export.

Payment terms

No.	Condition	Share	Amount (THB)
1	On signing the contract	40%	–
2	When the panels, inverter and batteries are delivered to site	50%	–
3	After testing and commissioning are complete	10%	–

Warranty and conditions

1. Warranty: panels, inverter and batteries under each manufacturer’s warranty; installation workmanship 2 years by Solar ESS.
2. Completion within 45 working days from contract signing and receipt of the first instalment in full (rain days and days off not counted). The customer makes the site ready for work to start.
3. Includes the PEA grid-connection application with an SLD certified by an electrical engineer. PEA fees, if any, are not included.
4. Roof structure is inspected before installation. Strengthening the roof, replacing damaged roof sheets or upgrading the existing MDB is quoted separately if needed.
5. Solar ESS supplies the energy from the solar system up to the connection point. Faults in the existing electrical system inside the building are outside Solar ESS’s responsibility.
6. Monitoring through the inverter app with automatic alerts. A 24-hour monitoring service is not included.
7. Copper cables, certified protection devices and certified electricians are used throughout. Prices are valid for 30 days.

Ask Solar ESS for a site survey: we confirm the roof, the 3-phase supply, the backed-up circuits and the final price for your site.

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